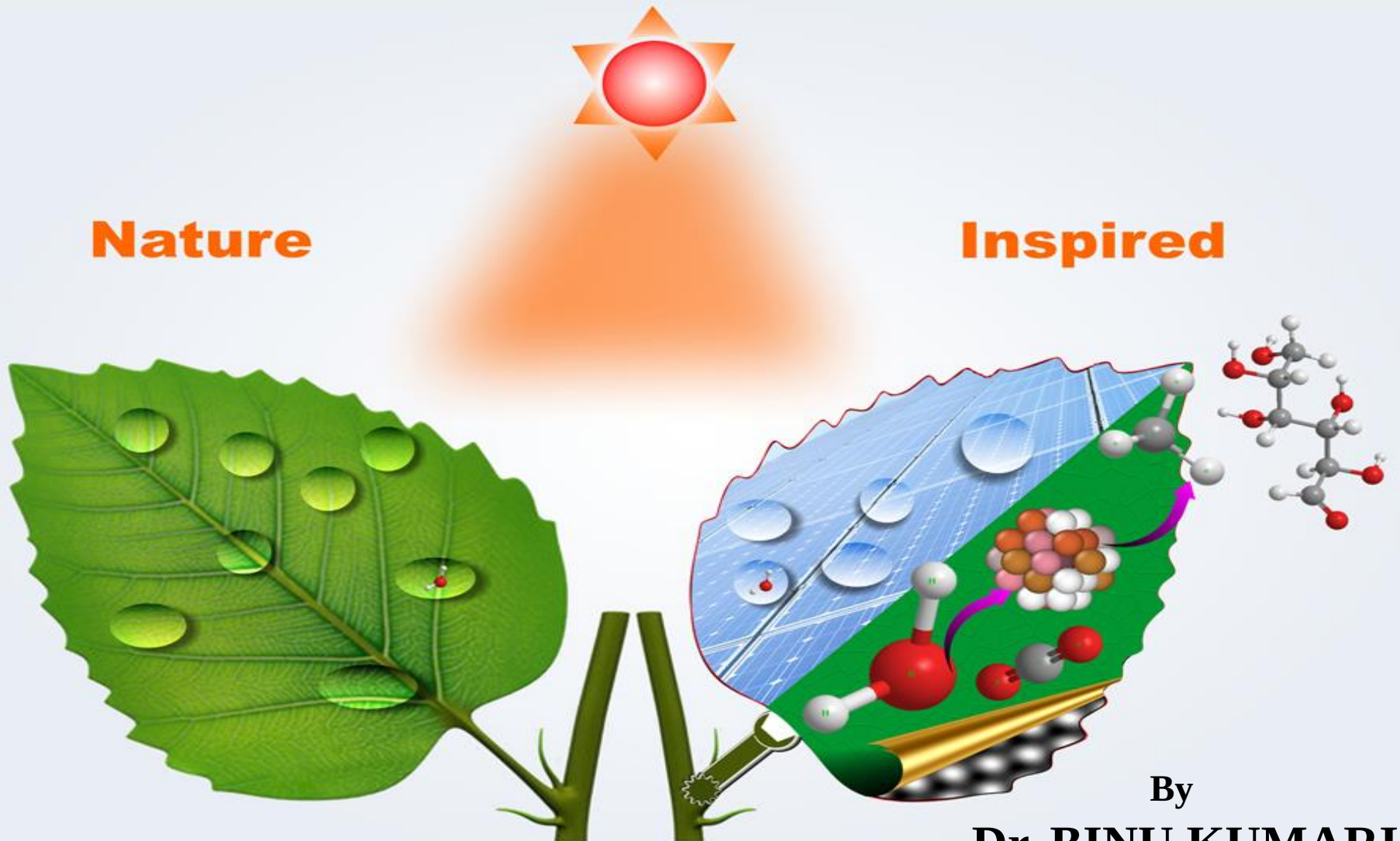


BIOLOGICALLY INSPIRED TECHNOLOGY AND ITS APPLICATION

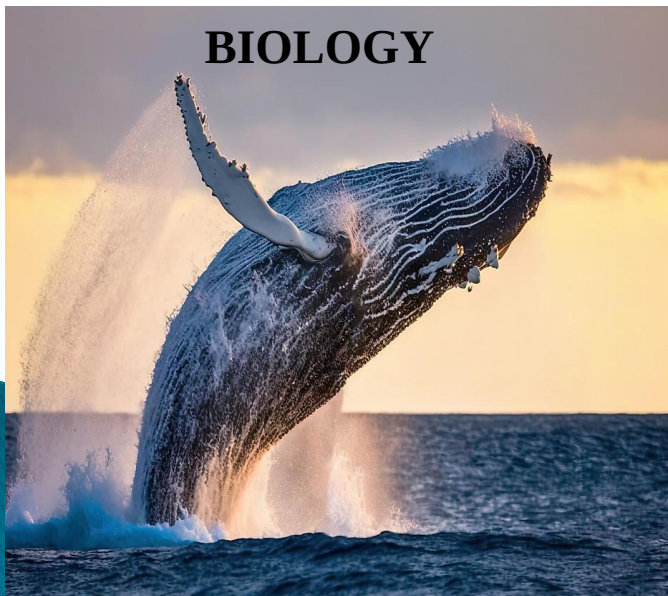


By
Dr. BINU KUMARI

NEED FOR BIOLOGY

- The engineering need to be exposed to the basics of biology so that **they would able to implement their ideas for solving various problems in biology and in our life.**
- **To address real life challenges and for industrial innovations to provide the needs of society.**
- **Artificial leaf to generate heat energy and electrical energy from solar energy.**
- All of us know that the airplanes were inspired by a bird flying.

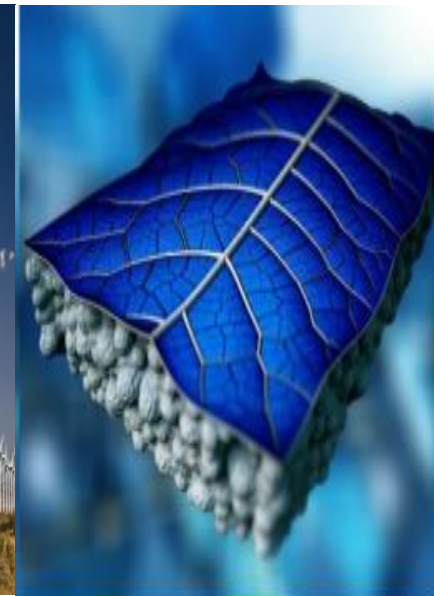
BIOLOGY



Engineer



Innovation



How Birds inspired Flight



-  Inspired by: Birds
-  Where: Italy & America
-  Innovation / Function: Human flight
-  Who: Leonardo da Vinci & the Wright Brothers (shared credit)

- Leonardo da Vinci applied biomimicry to the study of birds in the hope of enabling human flight.
- He very closely observed the anatomy and flight of birds and made countless sketches of proposed "flying machines"... known today as an aeroplane.
- These two American (Leonardo da Vinci and Wright brothers) **invented, built,** and flew the world's first successful **motor-operated aero plane.**

- The sweeping and elongated wings of soaring birds like **eagles and albatrosses** (their ability to fly without flapping their wings) **have influenced the design of long-range airplanes.**



- These wings shapes optimize lift and **enabling airplanes to cover vast distances with minimal effort.**

- The short and pointed wings of birds like **falcons and hawks** have inspired fighter jet designs (US).



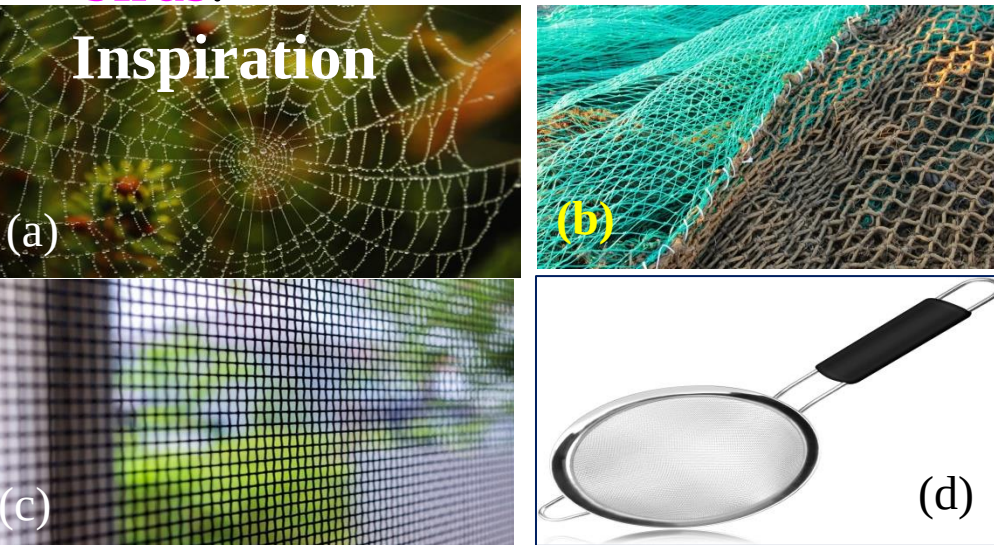
- These wings allow for quick moving and high-speed flight, essential for aerial combat (fight) and unexpected event that can occur at anytime, anyplace.

- ❑ It was not an easy task, but ultimately, mankind got there, and nowadays it's a very standard form of travel that we take for granted.
- ❑ **Thus the use of engineering to model biological systems has made our lives easier and understandable.**

Biologically inspired human innovation

- The presence of spiders in human habitats should have had some inspiring role in making such things as wires, ropes, nets, sieves, screens and woven fabrics.
- The fishing net, the screen in the screen doors, kitchen sieve etc. all is inspired by spider's web.
- In addition to the sieve, another inspired tool for application to food handling is the tong, which was probably inspired by the beak of birds.

Inspiration



Inspiration



Application



Four legs of most animals have been an inspiring model for furniture

Fig: The spider web has probably been inspiration for various human made tools (a) spider web, (b) Net, (c) screen door, (d) kitchen sieve

Case study

Shinkansen: The bullet train



Shinkansen: The bullet train and tunnel sonic booms

- ▶ Japanese engineers modeled the **Shinkansen a high speed bullet train of Japan**, after Kingfisher birds to solve one of their biggest problems... **tunnel sonic booms**.
- ▶ **Shinkansen bullet train speed was increased from 170 to 270 km/h**, **it** causes atmospheric pressure waves to build up in front of the train as it passes through tunnels **and** disturbing thousands of nearby residents as it emerges from tunnels.
- ▶ This compression wave propagates through the tunnel at the speed of sound, forming a micro-pressure wave, or a **tunnel sonic boom** at the exit.
- ▶ Shinkansen trains ran through many tunnels in dense neighborhoods, and the sonic booms were so forceful that residents could hear them 400 meters away.
- ▶ Thousands of residents were disturbed.

- To address the sonic boom problem a team was assembled.
- An engineer **Eiji Nakatsu** said that train causes sonic booms in tunnels **due to a sudden change in air resistance**.
- Nakatsu wasn't only an engineer, he was also an bird watcher.



- The bird Kingfishers can dive from the air (low resistance) to water (high resistance) at speeds of up to 25 mph without making a splash and catch a fish.
- Kingfishers could dive like that because of the streamlined shape of its long, pointed, wedge-shaped beak.





MASTER OF THE DIVE

- With an idea in mind, the team of engineers redesigned **Shinkansen's with a pointed front** (Head of train) similar to a **kingfisher's beak**.



Fig: The new Shinkansen design compared to a kingfisher

- Pantographs, the part that connect the train to electricity supply lines above was designed based on owl's serrated feathers.

Advantages

- The redesigned train was not only acceptably quieter when it exited the tunnels in dense residential areas.
- It was also 10% faster and 15 % more energy efficient.

BIOLOGICAL INSPIRED TECHNOLOGIES

- ❑ How the Fungi inspired local land restoration and toxic waste cleaning.
- ❑ How the humpbacks fins inspired more efficient wind turbines.
- ❑ How Mosquito proboscis inspired a less painful needle .



How the Fungi inspired local land restoration and toxic waste cleaning



Soil pollution

- ▶ **Disposal of highly toxic industrial or chemical waste** destroying the soil quality and toxifying the land.
- ▶ Industries such as Pharmaceutical, agricultural, dye & pigment, leather tanning, mining, textile, oil and refinery, chemical and refractory causes **soil and water pollution** due to the disposal of industrial waste, heavy metals, toxic chemicals, dumping oil and fuel, etc.
- High concentrations of heavy metals produced from different industries can accumulate and be transferred up the food chain into humans.
- When it is consumed in higher quantity, it can be toxic and affecting the function of the brain, lungs, kidney, and other important organs.

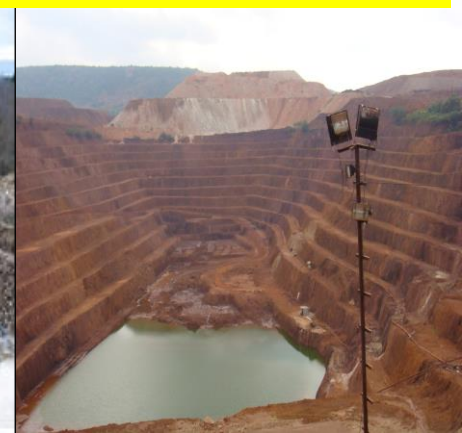
Soil contamination due to contaminant discharge from different industries



Food and agricultural industry



Industrial pollution due to mining

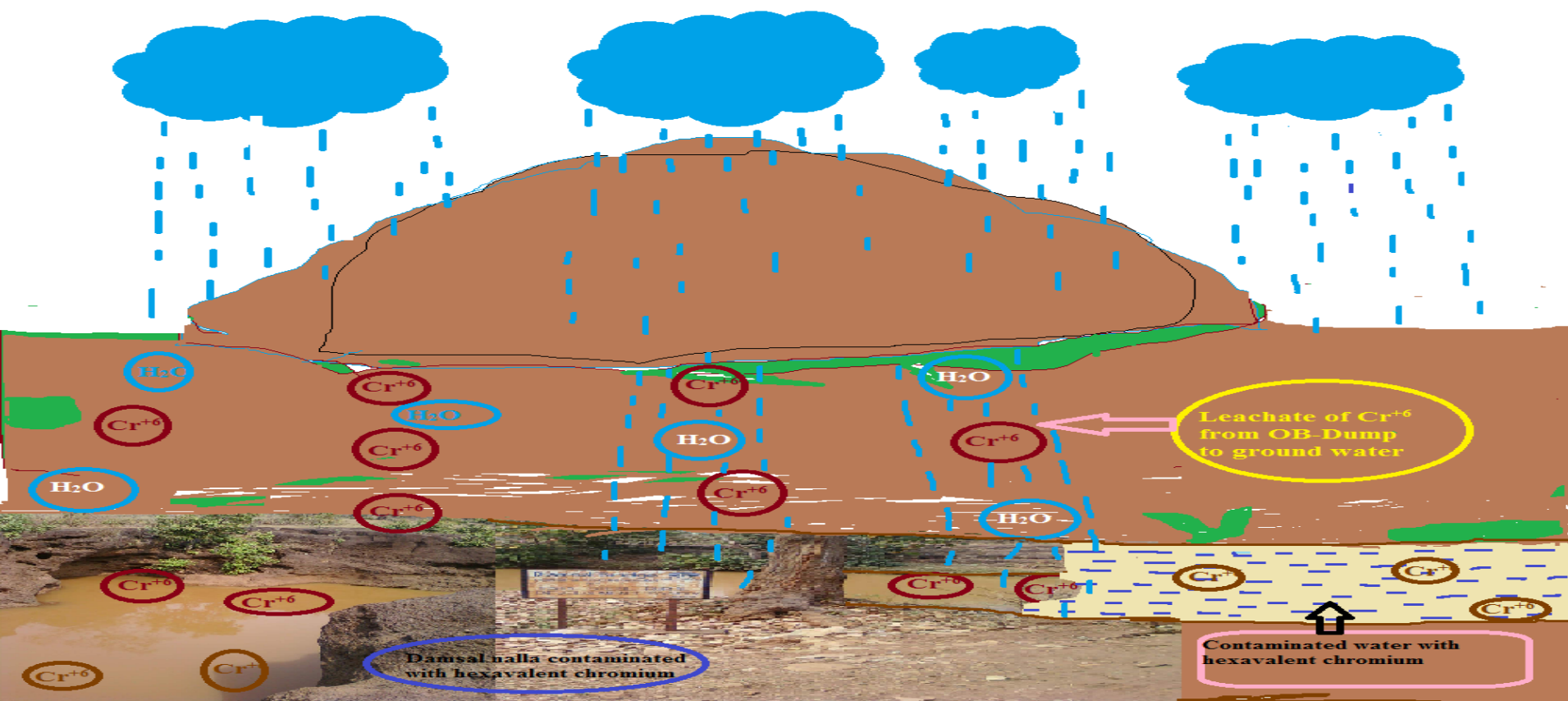


Coal mines



Sukinda Chromite mines, Odisha





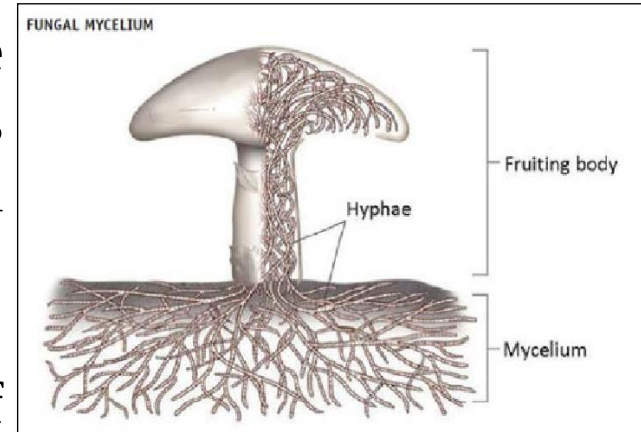
Soil remediation

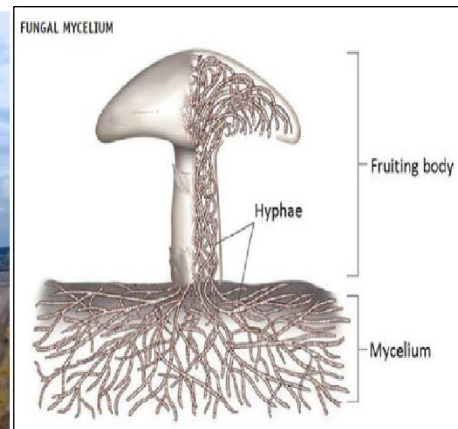
- Soil remediation is restoring or improving the **quality of soil that has been contaminated** or degraded by various pollutants, such as heavy metals, pesticides, petroleum products and industrial chemicals etc.

Importance of soil remediation

1. **Environmental Protection:** Contaminated soil can have adverse effects on the environment. It can lead to **soil erosion, groundwater pollution, and harm to local ecosystems.**
2. **Agricultural Productivity:** Healthy soil is essential for agriculture, as it provides the necessary nutrients and support for plant growth. **Soil contamination** can lead to **reduced crop yields and food safety concerns.** Remediated soil can restore fertile ground for farming, ensuring food security and quality.
3. **Human Health:** Contaminated soil can pose serious health risks to humans, especially **if the contaminants leach into the water supply** or are taken up by plants in the food chain.
4. **Land Reclamation:** Remediated soil can be used for various purpose including land for residential, commercial, and recreational purposes.

- **Soil restoration** is the process of improving the soil porosity, microbial life (add Soil Microbes to Restore Soil Health), nutrient density, and overall carbon levels of soil.
- **Bioremediation** is a branch of biotechnology that employs the use of living organisms, like **microbes and bacteria**, in the removal of contaminants, pollutants, and toxins from soil, water, and other environments.
- **Mycoremediation** is a form of bioremediation that employs fungi (the thread-like vegetative part of fungi) to break down or absorb contaminants in the soil, restore and heal contaminated ecosystems.
- Myco-, means fungi, and ‘remediate’, refers to restoring by reversing or stopping environmental damage.



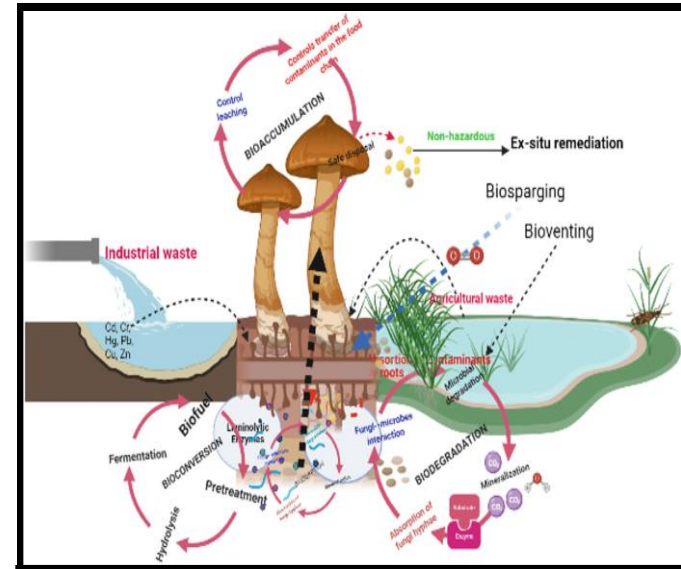


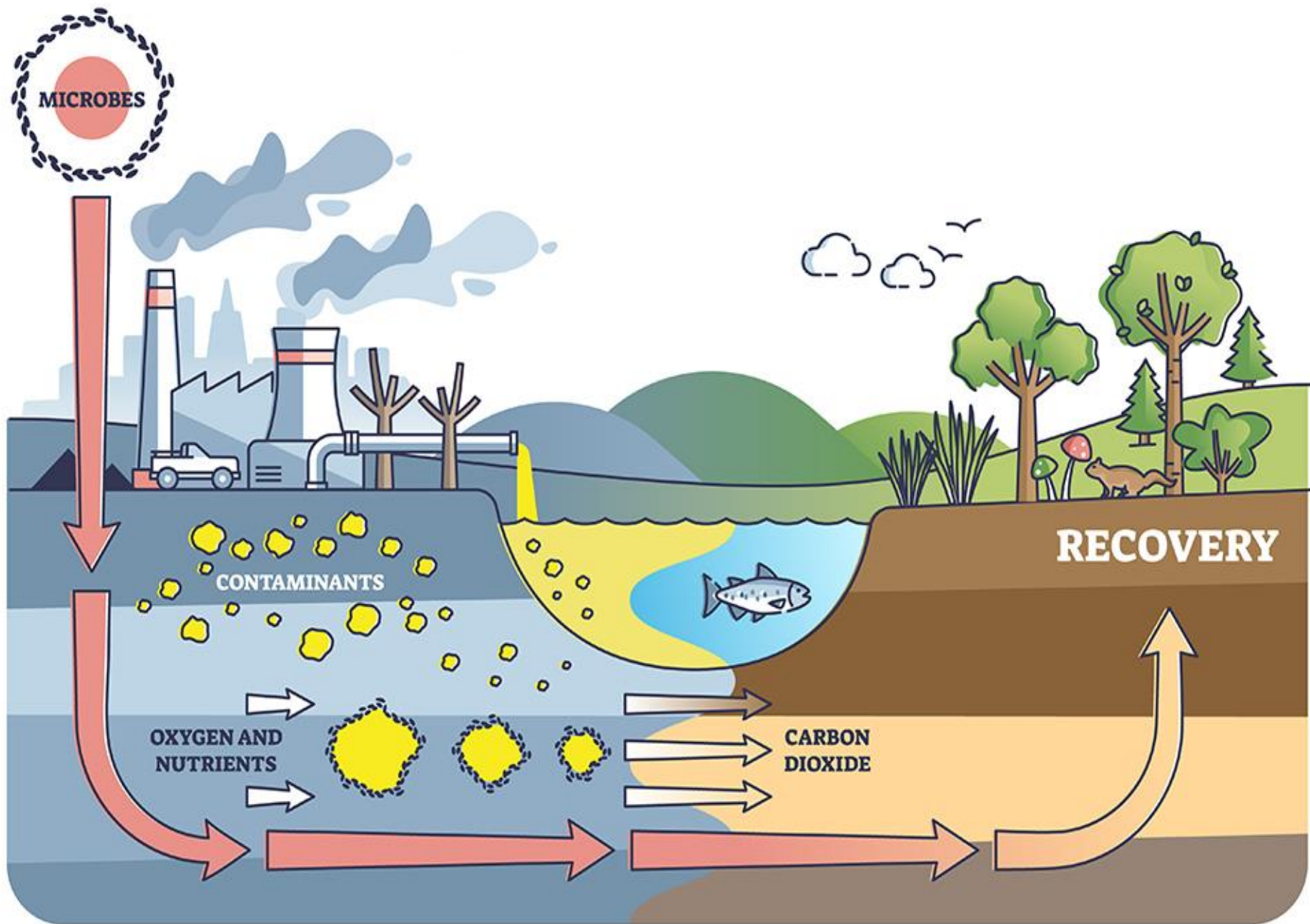
- ▶ Fungi, particularly **the mycelium**, the vegetative part of the fungus, possess the unique ability to grow in contaminated soils where other organisms struggle.
- ▶ As they grow, **they release enzymes that break down toxic pollutants, transforming them into harmless byproducts.**
- ▶ Researchers at **Novobiom** have turned the field of bioremediation on its head by locating fungi already in stages of reducing synthetics materials **into harmless substituents and selectively breeding them to treat heavily polluted waste.**
- ▶ All this can be done locally **without hauling the soil to any other location for treatment.**
- ▶ If we train fungi to treat the waste that we have already created, it can go a long way to converting our planet to its pre-industrial and agricultural glory.

Why mycelium ?

Mycelium has several properties that make it effective in soil remediation:

- 1. Biodegradation:** Mycelium **can secrete enzymes** that break down complex organic molecules, making them more easily metabolized by other microorganisms and **reducing the toxicity of contaminants**.
- 2. Metal Accumulation:** Some species of fungi have the ability to accumulate heavy metals in their mycelium. This can help to immobilize metals, preventing them from leaching into groundwater.
- 3. Soil Structure Improvement:** Mycelium can also improve soil structure by **binding soil particles together, increasing soil porosity, and enhancing water retention**.
- 4. Low Environmental Impact:** Mycoremediation is often considered an environmentally friendly approach because **it typically requires minimal external inputs and doesn't produce harmful byproducts**.





How the humpbacks whale fins inspired more efficient wind turbines



How the humpbacks fins inspired more efficient wind turbines



We plan to license the turbine retrofit elements within the next twelve months and to have retrofit leading edges for commercial wind turbines under development by mid-2014.

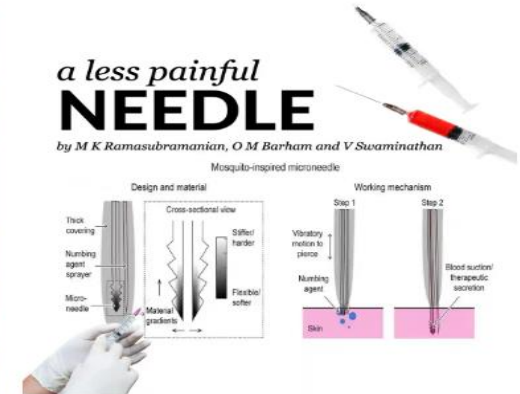
- Wind turbines is a device that converts the kinetic energy of wind into electrical energy.
- Inspired by the “bumpy” flippers of humpback whales, the team of Frank Fish, **innovated turbine blades** with irregular bumps (**tubercles**) on their leading edges.
- The humpback whale’s flippers with irregular bumps known as **tubercles**, might be responsible for adding extra efficiency.
- **Tubercles** on **turbines** blade give greater lift, **reduce drag** and **gathered more air** and **produces more energy from the winds**.

How Mosquito proboscis inspired a less painful needle

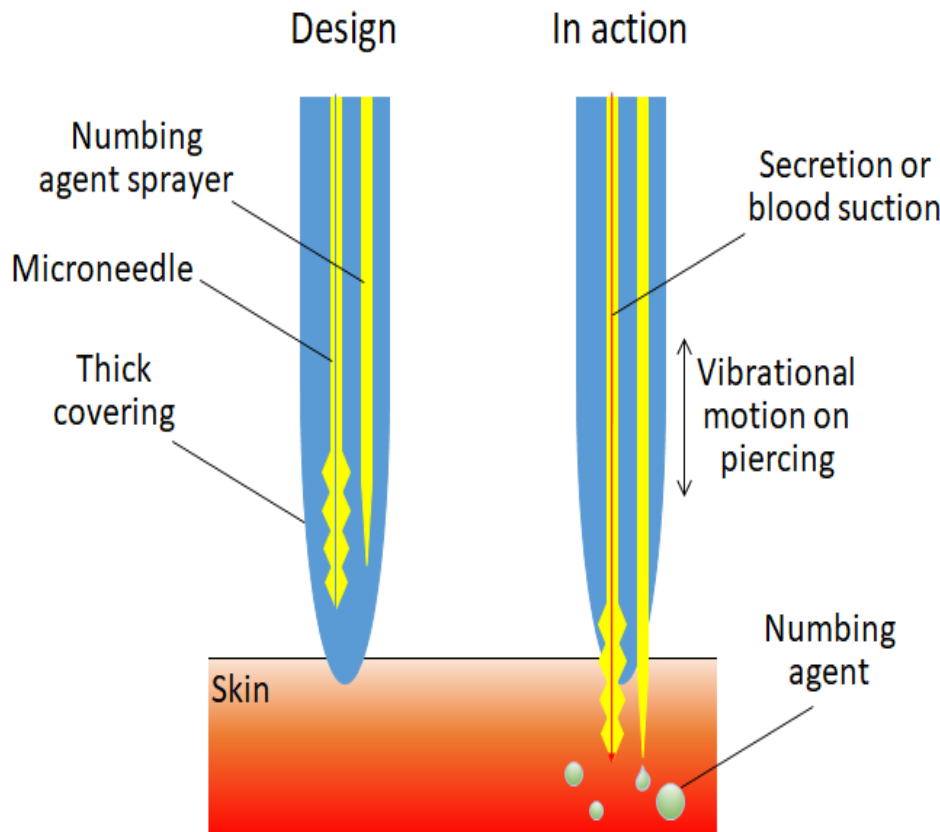


- Mosquito proboscis is made of multiple different needles each with its varied use so that the piercing is not noticeable at all.
- Research in 2008, mimicking the mosquito, **develop a 3 prong needle** that significantly reduced the pain caused by needle insertion.

How Mosquito Proboscis Inspired A Less Painful Needle



Inspired by: Mosquito proboscis
Where: India & Japan
Innovation/function: Less painful needle
Who: M K Ramasubramanian, O M Barham and V Swaminathan



- The mosquito punctures human skin to draw blood without the human even knowing it is happening. **They do this using three techniques.**
- **First**, upon insertion, **they secrete numbing saliva.**
- **Second**, the **fascicle**, or **part that draws blood, vibrates** when piercing the skin, reducing the force inflicted on the human.
- And third, the mosquito's “needle” is serrated, makes the insertion easier and less painful for the human.

Camera inspired by Eye

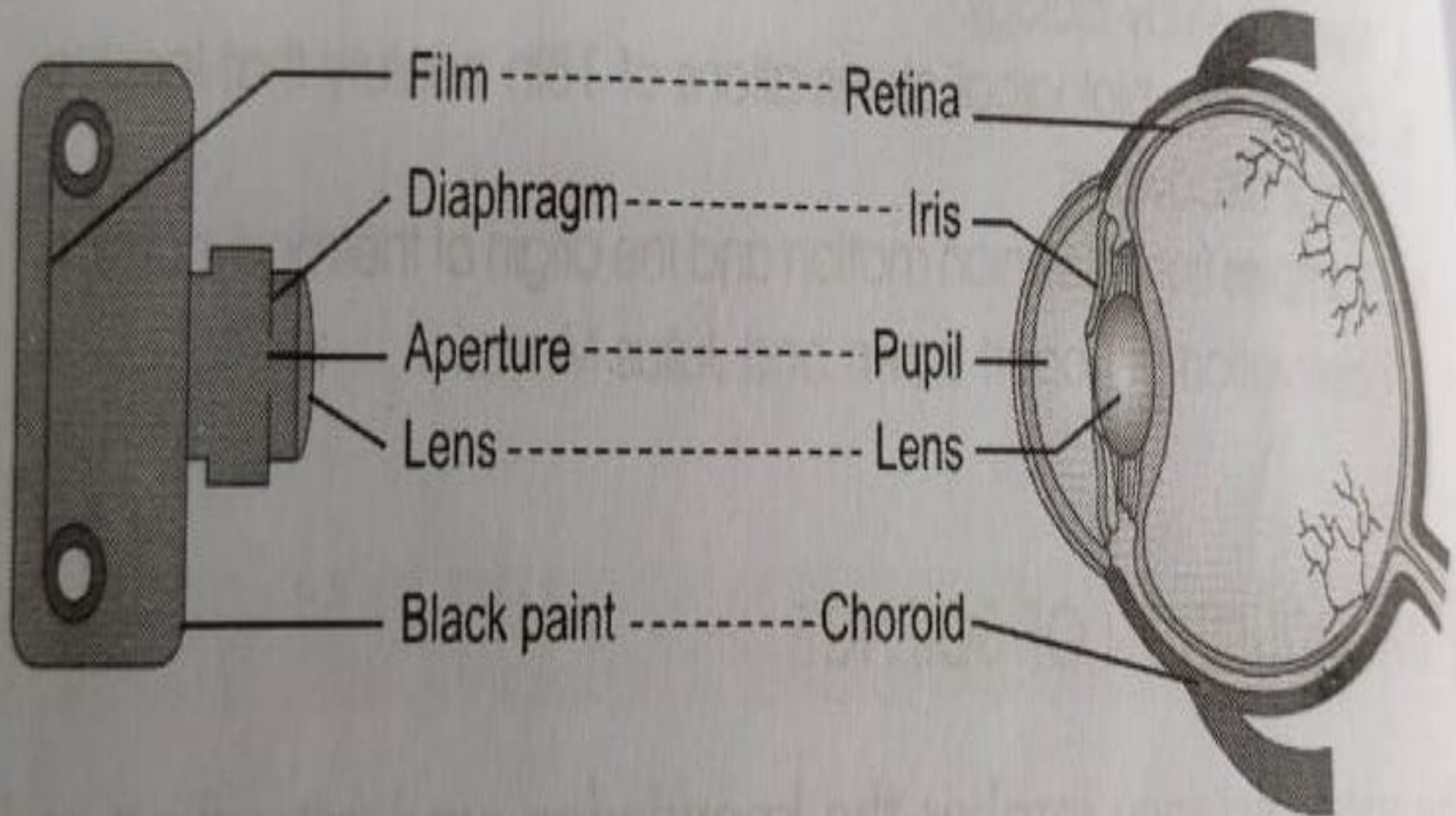


Fig. 1.1: Comparison between camera and human eye

Similarities between camera and human eye

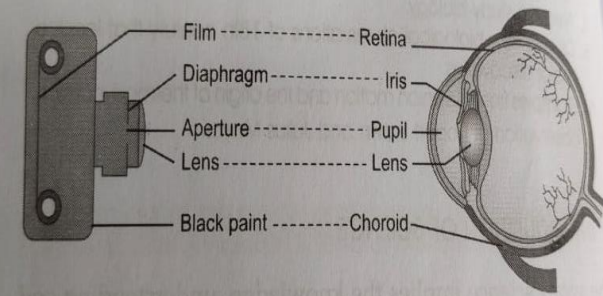


Fig. 1.1: Comparison between camera and human eye

Table 1.1 Similarities between camera and human eye

<i>Part of the camera</i>	<i>Corresponding part of an eye</i>	<i>Function</i>
Aperture	Pupil	Light enters the eye through the pupil/aperture
Diaphragm	Iris	The iris/diaphragm regulates the amount of light entering the eye/camera.
Lens	Lens	Focus light and image on the retina in eye and film in camera
Film	Retina	The part on which images are formed
Black paint	Choroid	The dark-colored melanin pigment in the choroid and black paint in camera absorbs light and limits reflections within the eye that could degrade vision

Differences between camera and human eye

Table 1.2 Differences between camera and human eye

<i>Camera</i>	<i>Human eye</i>
Focal length of lens is fixed	Focal length of the lens can be changed
Photographic film retains the image permanently	Retina retains the impression of an image for only $1/16^{\text{th}}$ second
A photograph has to be changed for getting next image	Same retina can be used for viewing unlimited images
Image is formed on photographic film and processing can be done through computer	Image is formed on retina which is further processed in brain